Computer Network LAB-01 Report :-

Experiment Title :-

Study of LAN Topology, Network Devices and Address configuration.

Objective :-

(i) To study the LAN network structure and topology used in the acedmic building.

(ii) To identify networking devices and observe their interconnection in the server room and classrooms.

(iii) To understand wired/wireless communication, IP configuration, MAC address and network data flow.

(iv) To ~~m~~measure network speed, and Determine MAC Address & IP address. & also subnet mask

Problem Statement-1 :-

- Draw the Network topology and clearly mention all components such as Router, switch, AP used in your college Acedmic. and end devices

Soln :- During the campus visit of Acedmic building for the project, I had notice following components:

(i) Internet connection through (ISP fiber) in server room.

(ii) Router ÷ Placed in server room on the rack.

(iii) Firewall & layer-3 core switch ÷ Also installe in server room for security & network management and routing.

(iv) Managed PoE + Switch ÷ Switches used to supply power and connectivity to network devices

(v) A P controller ÷ control all the wifi Access point of all the floors

(vi) Ethernet cable ÷ switch and patch panel connections, also used in classrooms.

(vii) floor Distribution cabinet ÷ wall mounted small switch cabinet on all the floors.

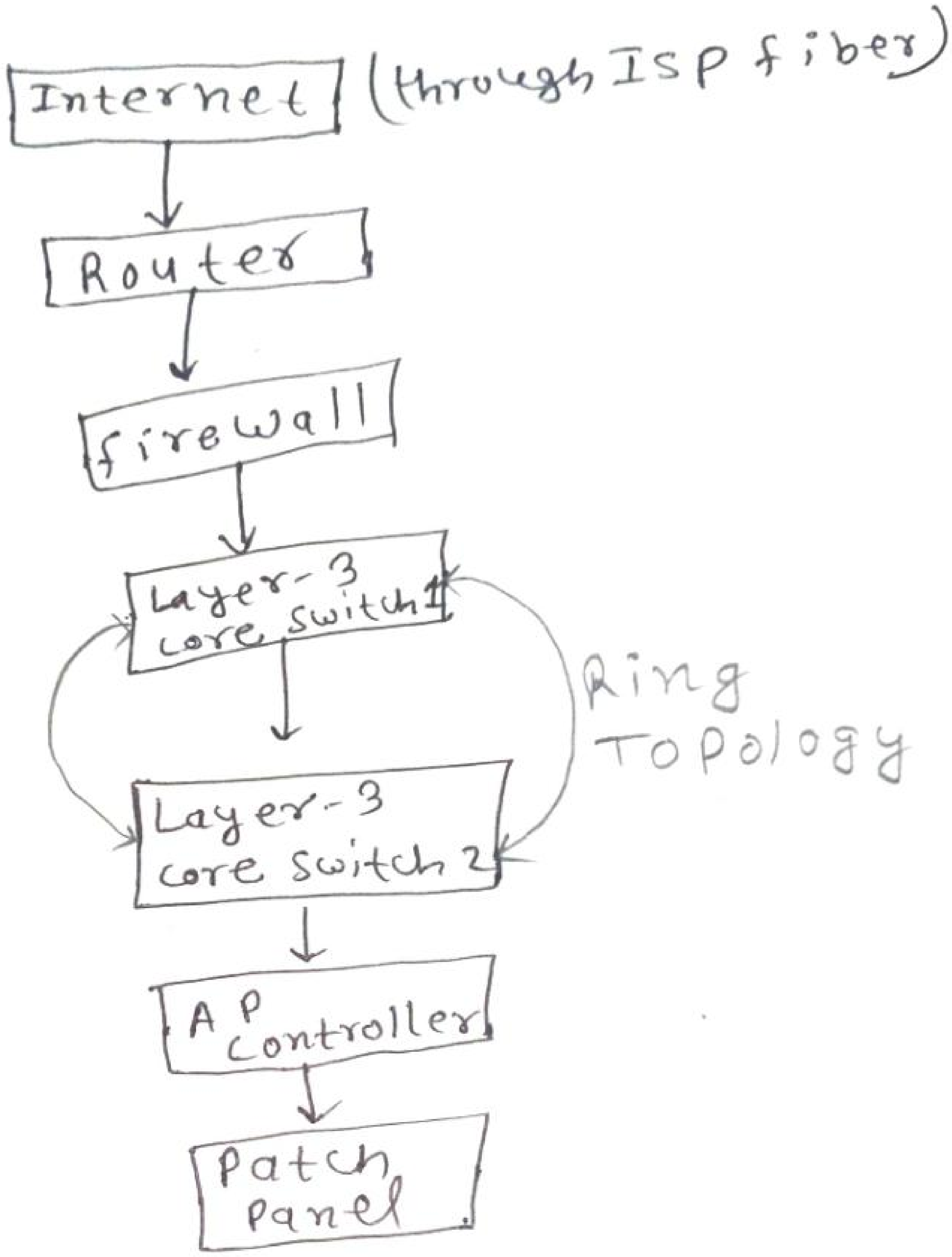
(viii) A P (Access Point) ÷ on the ceiling of every floor.

*The college campus network use a Hybrid Hierarchical design/~~where~~ Topology that combines:
- Tree Topology
- Extended star Topology
- Ring Topology (in server room)

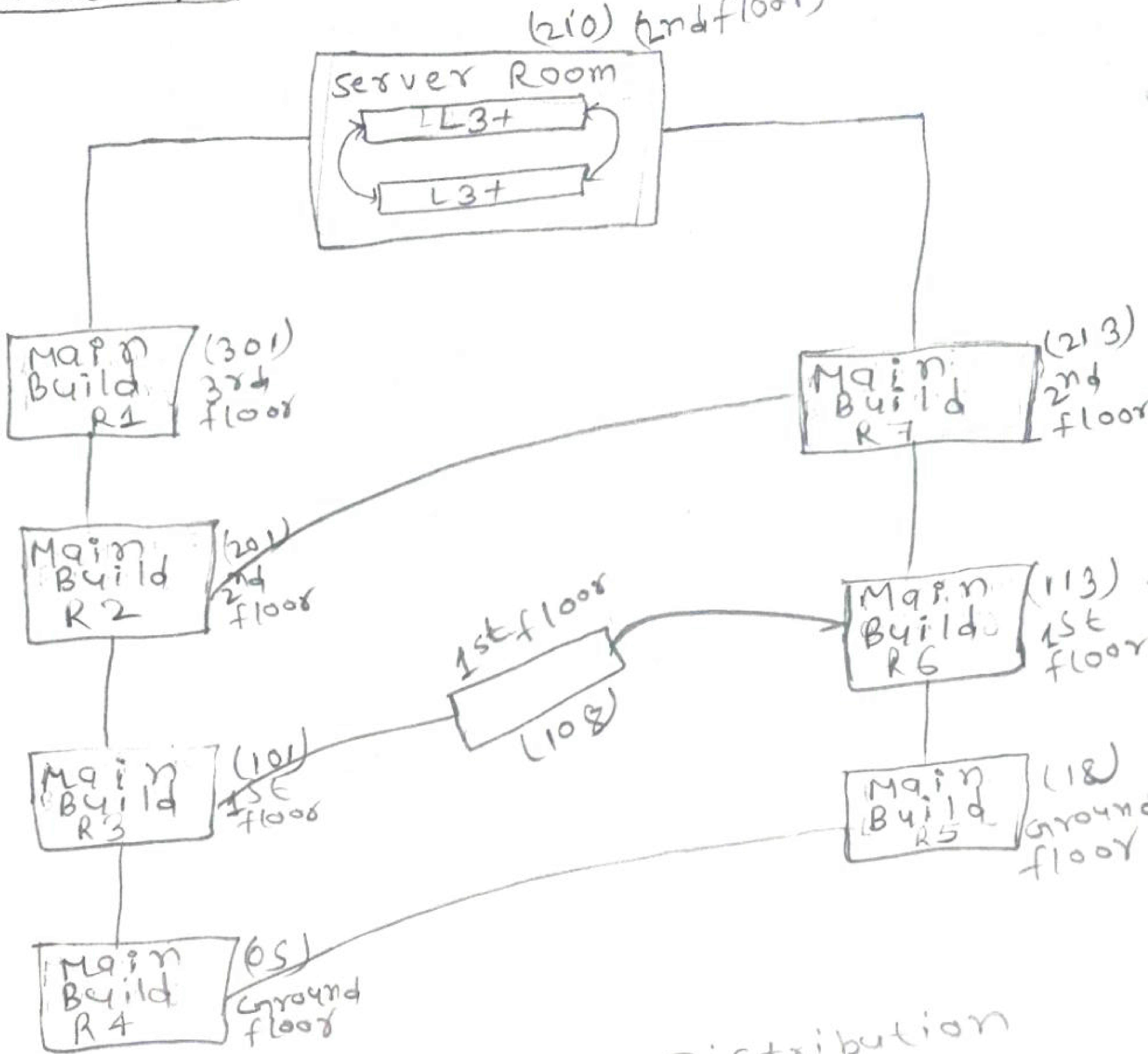
explanation ÷ The network infrastructure follows a hybrid hierarchical design where the server room acts as the central point. Core switch distribute connectivity to floor switches, access points, smart classrooms, labs, and

CCTV systems using fiber optic and Ethernet links. The backbone network also supports ring connectivity for redundancy and reliable communication.

Server Room Architecture/Topology :-



TOPOLOGY

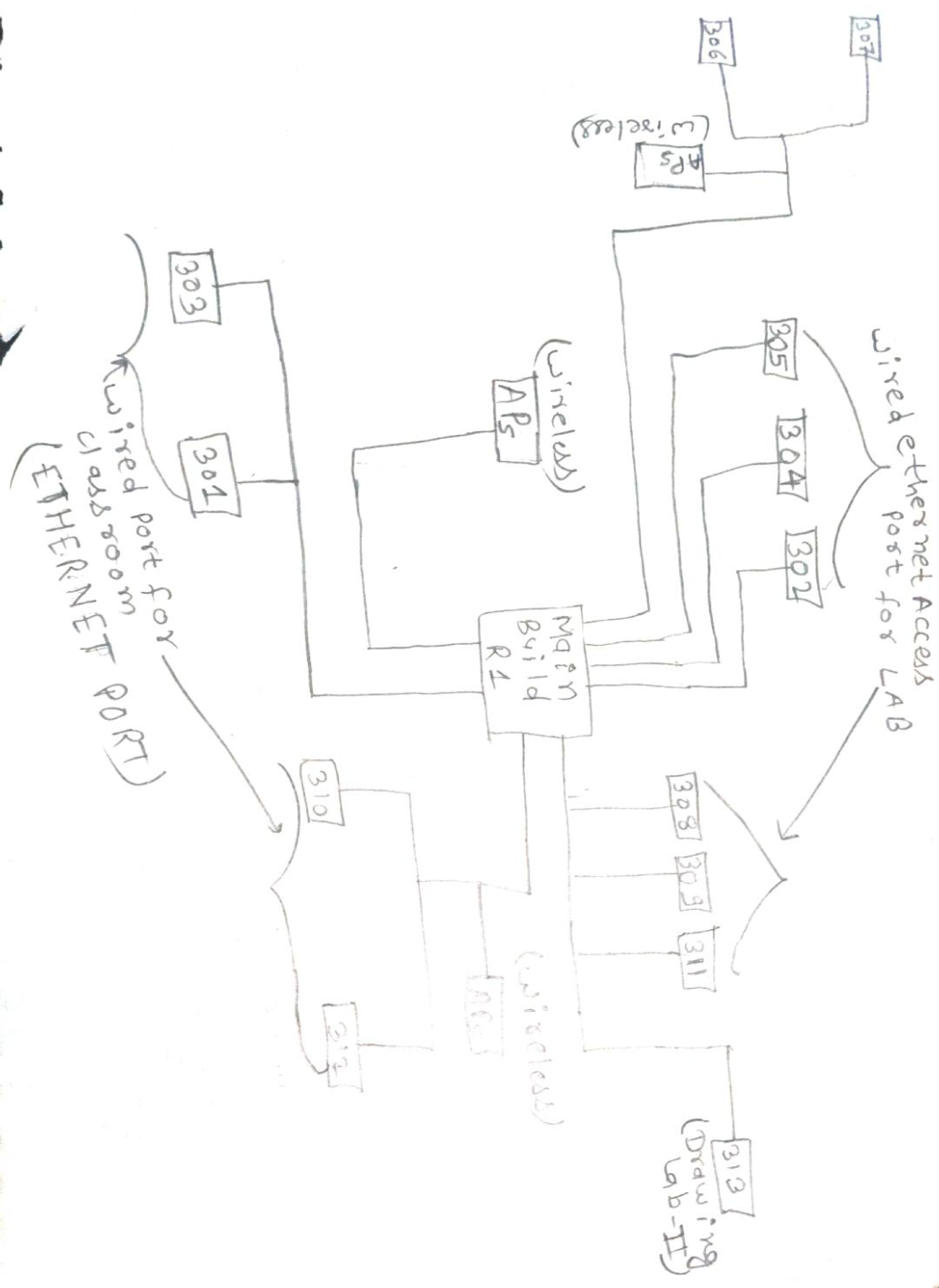


→ R1, R7, R2, R6 → Distribution layer

→ server room → core layer

→ R1, R2, R3, R4, R5, R6, R7 → access layer

Access layer switch (3rd floor-301) =



(2) Identify Network Devices ÷

⟨i.⟩ List all networking devices used in the LAN

(ii) Mention the function of each device.

• The following networking devices are used in the following LAN ÷

Device	function
• Router	connects the local network to the internet and manages routing.
• Layer-3 switch	perform switching and routing both
• PoE + switch	Provides power as well as data flow.
• firewall	protect from unauthorized access and cyber threat
• wireless Access point (AP)	Provide wifi connectivity to wireless users.
• server	stores and manages network resources and services.
• End Devices (PCs/laptops)	used by users to access network resource and internet services.

③ Analyse connection types :-

The college campus network uses both wired (Ethernet) and wireless (wi-fi) connections for communication.

1.) Wired connections are used in :-

(i) Computer lab
(ii) Server room
(iii) Desktop computers
(iv) Network switches and routers and APs

• Ethernet cable are used to provide stable and high speed communications.

2.) Wireless connection (wi-fi) used in :-

i.) Laptops
ii) Mobile phones
(iii) Smart boards in classroom
(iv) ~~class~~ Campus Areas

• Wi-fi Access point (APs) provide wireless internet connectivity across the campus.


Task Manager
Performance
Processes
Performance
App history
Startup apps
Users
Details
Services
Settings
CPU
Memory
Disk 0 (C: D:)
SSD (NVMe)
Ethernet
Wi-Fi
Wi-Fi
GPU 0
Wi-Fi
Intel(R) Wi-Fi 6 AX203
Send
16.0 Kbps
Receive
0 Kbps
Adapter name:
Signal strength:

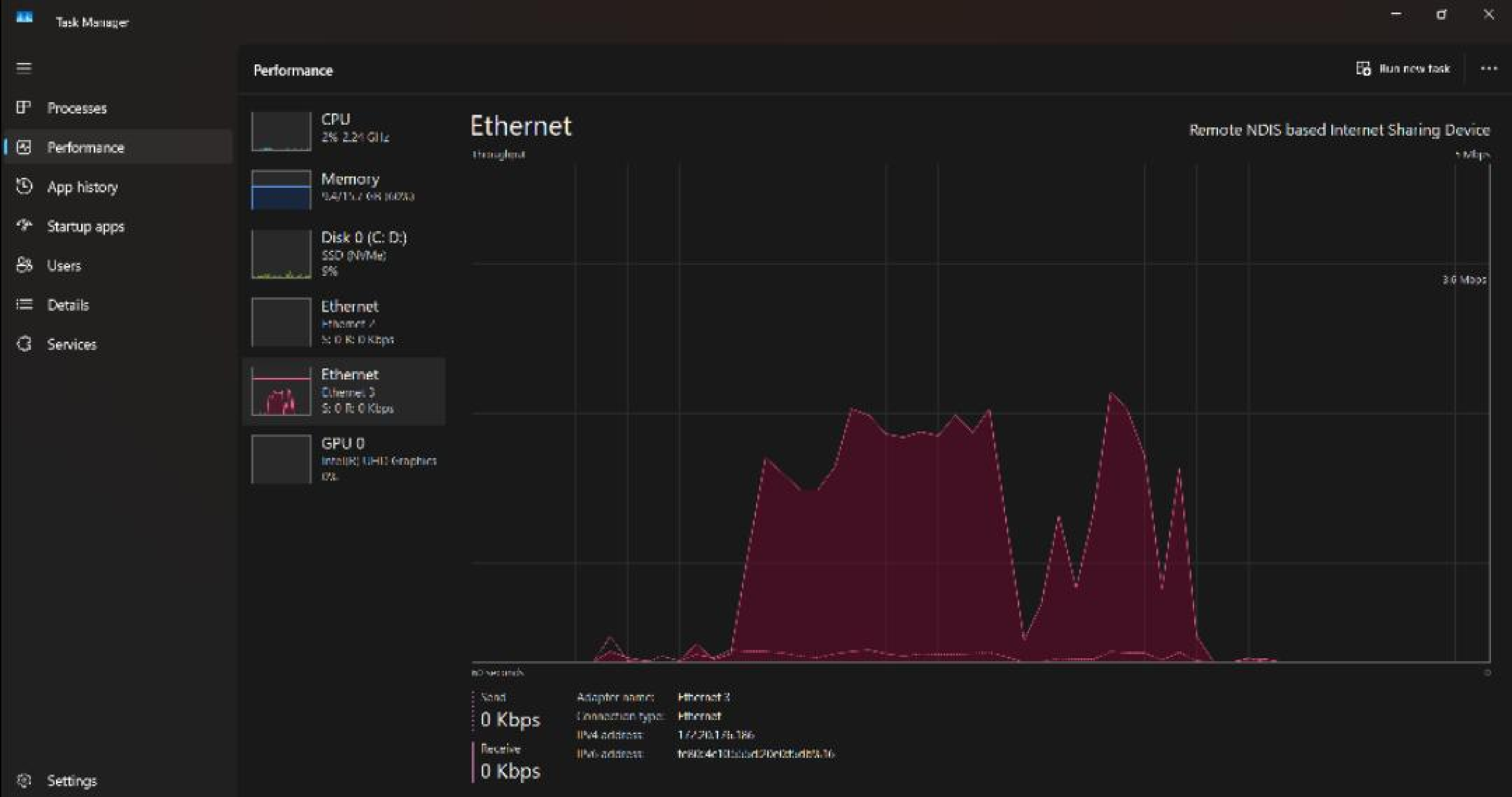
C:\Windows\system32\cmd.e:

```
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Abhishek>netsh wlan show interfaces

There is 1 interface on the system:

    Name                   : Wi-Fi
    Description            : Intel(R) Wi-Fi 6 AX203
    GUID                   : cd9b0e7f-d39f-43f0-a08f-f3b102c05540
    Physical address       : d6:7b:19:42:85:01
    Interface type         : Primary
    State                  : connected
    SSID                   : realme 8
    AP BSSID               : 4a:cf:4a:88:e9:6c
    Band                   : 2.4 GHz
    Channel                : 4
    Connected Akm-cipher   : [ akm = 00-0f-ac:02, cipher =  00-0f-ac:04 ]
    Network type           : Infrastructure
    Radio type             : 802.11n
    Authentication         : WPA2-Personal
    Cipher                 : CCMP
    Connection mode        : Profile
    Receive rate (Mbps)    : 65
    Transmit rate (Mbps)   : 65
    Signal                 : 97%
    Rssi                   : -33
    Profile                : realme 8
    QoS MSCS Configured         : 0
    QoS Map Configured          : 0
    QoS Map Allowed by Policy   : 0
```



Task Manager
Performance
Run new task
Processes
Performance
App history
Startup apps
Users
Details
Services
Settings
CPU
Memory
Disk 0 (C: D:)
SSD (NVMe)
Ethernet
S: 0 R: 0 Kbps
Ethernet
Ethernet 3
S: 0 R: 0 Kbps
GPU 0
Ethernet
Remote NDIS based Internet Sharing Device
Send
0 Kbps
Receive
0 Kbps
Adapter name:
Ethernet 3
Connection type:
Ethernet
IPv4 address:
IPv6 address:

```
C:\Users\Abhishek>netsh wlan show interfaces

There is 1 interface on the system:

    Name                   : Wi-Fi
    Description            : Intel(R) Wi-Fi 6 AX203
    GUID                   : cd9b0e7f-d39f-43f0-a08f-f3b102c05540
    Physical address       : b8:f7:75:71:de:fa
    Interface type         : Primary
    State                  : disconnected
    Radio status           : Hardware On
                             Software Off
```

```
C:\Users\Abhishek>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::3895:4076:6b87:249%20
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2409:408a:1e46:e0fd:5bb8:3d86:3a0b:c416
   Temporary IPv6 Address. . . . . . : 2409:408a:1e46:e0fd:98b1:3969:241f:fba
   Link-local IPv6 Address . . . . . : fe80::109:5381:6f00:f1cb%15
   IPv4 Address. . . . . . . . . . . : 10.97.225.50
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::403a:91ff:fec9:a94f%15
                                       10.97.225.225
```

```
C:\Users\Abhishek>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::3895:4076:6b87:249%20
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2409:408a:1e46:e0fd:6103:60a2:945d:11fb
   Temporary IPv6 Address. . . . . . : 2409:408a:1e46:e0fd:85a1:b4a5:9a24:da0d
   Link-local IPv6 Address . . . . . : fe80::67e2:b35b:88e2:146%16
   IPv4 Address. . . . . . . . . . . : 10.183.20.109
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::c4ac:55ff:fe6e:f1e3%16
                                       10.183.20.179
```

```
C:\Users\Abhishek>getmac

Physical Address    Transport Name
=================== ==========================================================
D6-7B-19-42-85-01   \Device\Tcpip_{CD9B0E7F-D39F-43F0-A08F-F3B102C05540}
0A-00-27-00-00-14   \Device\Tcpip_{F3C933FE-1979-4CBC-9D52-1EA6C2F1B46D}

C:\Users\Abhishek>
```

(6)(i) Difference b/w wired and wireless speed:-

These are various difference b/w wired and wireless speed:-

Wired Network	Wireless Network
→ uses Ethernet cable for communication	→ uses wifi/radio signal
→ Generally provides higher speed	→ speed may fluctuate
→ More stable and reliable	→ Affected by signal interference.
→ lower latency	→ Higher latency
→ Better for gaming and large file transfer	→ Better for mobility and convenience
→ less affected by distance	→ speed decreases with distance.

(ii) Role of subnet mask in networking:-

Ans:- A subnet mask is used to divide an IP address into two parts:-

(i) Network ID

(ii) Host ID

Note:- A subnet mask is a 32-bit number used in computer networking to divide an IP address.

• It helps computers and networking devices identify whether the destination device belongs to the same or different network.

* functions of Subnet Mask :-

(i) Identifies Network and Host :- subnet mask seprates the network portion of an IP address for proper communication.

(ii) Helps in Data Routing :- It helps routers and switch determine the correct path for sending data packets between networks.

(iii) Reduces Network Traffic :- By dividing large networks into smaller subnet works, subnet masks reduce unnecessary traffic and improve network performance.

(iv) Enhances Security :- Different departments or systems can be separated into different subnetworks for better security and controlled access.

Example :- IP Address: 192.168.1.10

Subnet mask : 255.255.255.0

Here, .192.168.1 represents the network part
.10 represents the host /device part.